

1 Problem Setup & Static Representations

Before beginning the matrix projections, we define the starting elements of the translation system.

1.1 System Vocabulary and English Sentence

We have a vocabulary $\mathcal{V} = \{\text{cat}, \text{dog}, \text{fish}, \text{likes}\}$ with an embedding dimension of $d_{\text{model}} = 3$. The input English sentence to the encoder is:

“cat likes fish”

1.2 Static Word Embeddings

Each word is mapped to a static dense embedding vector:

$$\begin{aligned}\mathbf{e}_{\text{cat}} &= [1 \quad 0 \quad 1] \\ \mathbf{e}_{\text{likes}} &= [1 \quad 1 \quad 0] \\ \mathbf{e}_{\text{fish}} &= [0 \quad 1 \quad 1]\end{aligned}$$

Stacking these vectors gives our word embedding matrix X_{word} :

$$X_{\text{word}} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad (1)$$

1.3 Calculation of Positional Encodings (PE)

To make the model aware of sequence order, we calculate sinusoidal Positional Encoding vectors for the three positions ($\text{pos} = 0, 1, 2$) across our three dimensions ($j \in \{0, 1, 2\}$):

- **Position 0 (“cat”):**

$$PE_{\text{pos}=0} = [\sin(0) \quad \cos(0) \quad \sin(0)] = [0.0000 \quad 1.0000 \quad 0.0000] \quad (2)$$

- **Position 1 (“likes”):**

$$PE(1, 0) = \sin(1) \approx 0.8415$$

$$PE(1, 1) = \cos(1) \approx 0.5403$$

$$PE(1, 2) = \sin\left(\frac{1}{10000^{2/3}}\right) \approx \sin(0.002154) \approx 0.0022$$

$$PE_{\text{pos}=1} \approx [0.8415 \quad 0.5403 \quad 0.0022]$$

- **Position 2 (“fish”):**

$$PE(2, 0) = \sin(2) \approx 0.9093$$

$$PE(2, 1) = \cos(2) \approx -0.4161$$

$$PE(2, 2) = \sin\left(\frac{2}{10000^{2/3}}\right) \approx \sin(0.004309) \approx 0.0043$$

$$PE_{\text{pos}=2} \approx [0.9093 \quad -0.4161 \quad 0.0043]$$

Adding the positional coordinates to our static word embeddings yields the final input matrix X :

$$X = X_{\text{word}} + PE = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 \\ 1.8415 & 1.5403 & 0.0022 \\ 0.9093 & 0.5839 & 1.0043 \end{bmatrix} \quad (3)$$

2 Detailed Simulation of Problem 1 (Encoder Self-Attention)

Using the input matrix X , we simulate the entire internal calculations of the Encoder Self-Attention layer.

2.1 Projections to Query, Key, and Value Matrices

We are given the learnable weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{3 \times 3}$:

$$W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad W_K = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad W_V = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad (4)$$

2.1.1 1. Query Matrix calculation ($Q = X \cdot W_Q$)

$$Q = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 \\ 1.8415 & 1.5403 & 0.0022 \\ 0.9093 & 0.5839 & 1.0043 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.0000 & 2.0000 & 2.0000 \\ 1.8436 & 1.5425 & 3.3818 \\ 1.9136 & 1.5882 & 1.4932 \end{bmatrix} \quad (5)$$

2.1.2 2. Key Matrix calculation ($K = X \cdot W_K$)

$$K = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 \\ 1.8415 & 1.5403 & 0.0022 \\ 0.9093 & 0.5839 & 1.0043 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.0000 & 2.0000 & 2.0000 \\ 1.5425 & 1.8436 & 3.3818 \\ 1.5882 & 1.9136 & 1.4932 \end{bmatrix} \quad (6)$$

2.1.3 3. Value Matrix calculation ($V = X \cdot W_V$)

$$V = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 \\ 1.8415 & 1.5403 & 0.0022 \\ 0.9093 & 0.5839 & 1.0043 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2.0000 & 2.0000 & 2.0000 \\ 1.5425 & 3.3818 & 1.8436 \\ 1.5882 & 1.4932 & 1.9136 \end{bmatrix} \quad (7)$$

2.2 Dot-Product Similarity Scores ($Q \cdot K^T$)

Next, we calculate the similarity score matrix:

$$Q \cdot K^T = \begin{bmatrix} 2.0000 & 2.0000 & 2.0000 \\ 1.8436 & 1.5425 & 3.3818 \\ 1.9136 & 1.5882 & 1.4932 \end{bmatrix} \cdot \begin{bmatrix} 2.0000 & 1.5425 & 1.5882 \\ 2.0000 & 1.8436 & 1.9136 \\ 2.0000 & 3.3818 & 1.4932 \end{bmatrix} = \begin{bmatrix} 12.0000 & 13.5357 & 9.9898 \\ 13.5357 & 17.1238 & 10.9291 \\ 9.9898 & 10.9291 & 8.3077 \end{bmatrix} \quad (8)$$

2.3 Scaling by Key Dimension ($\frac{Q \cdot K^T}{\sqrt{3}}$)

We scale the attention scores by dividing each element by $\sqrt{d_k} = \sqrt{3} \approx 1.73205$:

$$\text{Scaled Scores} = \begin{bmatrix} 6.9282 & 7.8148 & 5.7676 \\ 7.8148 & 9.8864 & 6.3099 \\ 5.7676 & 6.3099 & 4.7965 \end{bmatrix} \quad (9)$$

2.4 Row-wise Softmax Transformation (α)

For each row, we exponentiate the elements and divide by the sum of that row to normalize the weights:

- **Row 0 ("cat"):** Sum of exponents: $e^{6.9282} + e^{7.8148} + e^{5.7676} = 1020.66 + 2477.01 + 319.78 = 3817.45$

$$\alpha_{\text{row } 0} = \begin{bmatrix} \frac{1020.66}{3817.45} & \frac{2477.01}{3817.45} & \frac{319.78}{3817.45} \end{bmatrix} = \begin{bmatrix} 0.2674 & 0.6489 & 0.0838 \end{bmatrix} \quad (10)$$

- **Row 1 (“likes”)**: Sum of exponents: $e^{7.8148} + e^{9.8864} + e^{6.3099} = 2477.01 + 19661.12 + 549.99 = 22688.12$

$$\alpha_{\text{row } 1} = \begin{bmatrix} \frac{2477.01}{22688.12} & \frac{19661.12}{22688.12} & \frac{549.99}{22688.12} \end{bmatrix} = \begin{bmatrix} 0.1092 & 0.8666 & 0.0242 \end{bmatrix} \quad (11)$$

- **Row 2 (“fish”)**: Sum of exponents: $e^{5.7676} + e^{6.3099} + e^{4.7965} = 319.78 + 549.99 + 121.08 = 990.85$

$$\alpha_{\text{row } 2} = \begin{bmatrix} \frac{319.78}{990.85} & \frac{549.99}{990.85} & \frac{121.08}{990.85} \end{bmatrix} = \begin{bmatrix} 0.3227 & 0.5551 & 0.1222 \end{bmatrix} \quad (12)$$

Thus, the global Attention Weight Matrix (α) is:

$$\alpha = \begin{bmatrix} 0.2674 & 0.6489 & 0.0838 \\ 0.1092 & 0.8666 & 0.0242 \\ 0.3227 & 0.5551 & 0.1222 \end{bmatrix} \quad (13)$$

2.5 Attention Output & Residual Addition

We calculate the weighted sum of Value matrix: $\text{Attn_Out} = \alpha \cdot V$:

$$\text{Attn_Out} = \begin{bmatrix} 0.2674 & 0.6489 & 0.0838 \\ 0.1092 & 0.8666 & 0.0242 \\ 0.3227 & 0.5551 & 0.1222 \end{bmatrix} \cdot \begin{bmatrix} 2.0000 & 2.0000 & 2.0000 \\ 1.5425 & 3.3818 & 1.8436 \\ 1.5882 & 1.4932 & 1.9136 \end{bmatrix} = \begin{bmatrix} 1.6686 & 2.8541 & 1.8913 \\ 1.5935 & 3.1851 & 1.8624 \\ 1.6957 & 2.7051 & 1.9026 \end{bmatrix} \quad (14)$$

Finally, we apply the residual addition to obtain our context-rich Encoder Output ($\text{enc} = X + \text{Attn_Out}$):

$$\text{enc} = \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 \\ 1.8415 & 1.5403 & 0.0022 \\ 0.9093 & 0.5839 & 1.0043 \end{bmatrix} + \begin{bmatrix} 1.6686 & 2.8541 & 1.8913 \\ 1.5935 & 3.1851 & 1.8624 \\ 1.6957 & 2.7051 & 1.9026 \end{bmatrix} = \begin{bmatrix} \mathbf{2.6686} & \mathbf{3.8541} & \mathbf{2.8913} \\ \mathbf{3.4350} & \mathbf{4.7254} & \mathbf{1.8645} \\ \mathbf{2.6050} & \mathbf{3.2890} & \mathbf{2.9070} \end{bmatrix} \quad (15)$$

3 Detailed Simulation of Problem 2 (Cross-Attention Memory)

Since the source sentence is static throughout decoding, we cache the final encoder outputs projected into Keys (K_{cross}) and Values (V_{cross}) to serve as frozen cross-attention memory.

3.1 Given Cross-Attention Weight Matrices

The cross-attention projection matrices provided in Question 2 are:

$$W_{K,\text{cross}} = \begin{bmatrix} 0 & 3 & 4 \\ 2 & 0 & 1 \\ 1 & 5 & 7 \end{bmatrix}, \quad W_{V,\text{cross}} = \begin{bmatrix} 0 & 1 & 7 \\ 6 & 4 & 0 \\ 8 & 0 & 3 \end{bmatrix} \quad (16)$$

3.2 Calculation of Cross-Attention Keys ($K_{\text{cross}} = \text{enc} \cdot W_{K,\text{cross}}$)

$$K_{\text{cross}} = \begin{bmatrix} 2.6686 & 3.8541 & 2.8913 \\ 3.4350 & 4.7254 & 1.8645 \\ 2.6050 & 3.2890 & 2.9070 \end{bmatrix} \cdot \begin{bmatrix} 0 & 3 & 4 \\ 2 & 0 & 1 \\ 1 & 5 & 7 \end{bmatrix} \quad (17)$$

3.2.1 Step-by-Step Row Multiplication

- **Row 0 (“cat” Key):**

$$K_{\text{cross},(0,0)} = 2.6686(0) + 3.8541(2) + 2.8913(1) = 10.5995$$

$$K_{\text{cross},(0,1)} = 2.6686(3) + 3.8541(0) + 2.8913(5) = 22.4623$$

$$K_{\text{cross},(0,2)} = 2.6686(4) + 3.8541(1) + 2.8913(7) = 34.7676$$

$$\text{Vector: } [10.5995 \quad 22.4623 \quad 34.7676]$$

- **Row 1 (“likes” Key):**

$$K_{\text{cross},(1,0)} = 3.4350(0) + 4.7254(2) + 1.8645(1) = 11.3153$$

$$K_{\text{cross},(1,1)} = 3.4350(3) + 4.7254(0) + 1.8645(5) = 19.6275$$

$$K_{\text{cross},(1,2)} = 3.4350(4) + 4.7254(1) + 1.8645(7) = 31.5169$$

$$\text{Vector: } [11.3153 \quad 19.6275 \quad 31.5169]$$

- **Row 2 (“fish” Key):**

$$K_{\text{cross},(2,0)} = 2.6050(0) + 3.2890(2) + 2.9070(1) = 9.4850$$

$$K_{\text{cross},(2,1)} = 2.6050(3) + 3.2890(0) + 2.9070(5) = 22.3500$$

$$K_{\text{cross},(2,2)} = 2.6050(4) + 3.2890(1) + 2.9070(7) = 34.0580$$

$$\text{Vector: } [9.4850 \quad 22.3500 \quad 34.0580]$$

3.3 Calculation of Cross-Attention Values ($V_{\text{cross}} = \text{enc} \cdot W_{V,\text{cross}}$)

$$V_{\text{cross}} = \begin{bmatrix} 2.6686 & 3.8541 & 2.8913 \\ 3.4350 & 4.7254 & 1.8645 \\ 2.6050 & 3.2890 & 2.9070 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 7 \\ 6 & 4 & 0 \\ 8 & 0 & 3 \end{bmatrix} \quad (18)$$

3.3.1 Step-by-Step Row Multiplication

- **Row 0 (“cat” Value):**

$$V_{\text{cross},(0,0)} = 2.6686(0) + 3.8541(6) + 2.8913(8) = 46.2550$$

$$V_{\text{cross},(0,1)} = 2.6686(1) + 3.8541(4) + 2.8913(0) = 18.0850$$

$$V_{\text{cross},(0,2)} = 2.6686(7) + 3.8541(0) + 2.8913(3) = 27.3541$$

$$\text{Vector: } [46.2550 \quad 18.0850 \quad 27.3541]$$

- **Row 1 (“likes” Value):**

$$V_{\text{cross},(1,0)} = 3.4350(0) + 4.7254(6) + 1.8645(8) = 43.2684$$

$$V_{\text{cross},(1,1)} = 3.4350(1) + 4.7254(4) + 1.8645(0) = 22.3366$$

$$V_{\text{cross},(1,2)} = 3.4350(7) + 4.7254(0) + 1.8645(3) = 29.6385$$

$$\text{Vector: } [43.2684 \quad 22.3366 \quad 29.6385]$$

- **Row 2 (“fish” Value):**

$$V_{\text{cross},(2,0)} = 2.6050(0) + 3.2890(6) + 2.9070(8) = 42.9900$$

$$V_{\text{cross},(2,1)} = 2.6050(1) + 3.2890(4) + 2.9070(0) = 15.7610$$

$$V_{\text{cross},(2,2)} = 2.6050(7) + 3.2890(0) + 2.9070(3) = 26.9560$$

$$\text{Vector: } [42.9900 \quad 15.7610 \quad 26.9560]$$

3.4 Final Cached Matrices

The finished cached memories representing the context of our source English phrase are:

$$K_{\text{cross}} \approx \begin{bmatrix} 10.5996 & 22.4623 & 34.7677 \\ 11.3154 & 19.6277 & 31.5172 \\ 9.4848 & 22.3498 & 34.0576 \end{bmatrix} \quad (19)$$

$$V_{\text{cross}} \approx \begin{bmatrix} 46.2552 & 18.0852 & 27.3542 \\ 43.2690 & 22.3367 & 29.6386 \\ 42.9891 & 15.7606 & 26.9559 \end{bmatrix} \quad (20)$$